

AMBIENT DISPLAY

Complete Product, Technical Architecture & Mobile Redesign Blueprint (Live Capture Edition)

1. Executive Product Brief

The Core Problem: Conventional alarm clocks force an unnatural, rigid schedule. If you plan to sleep for 7.5 hours and set an alarm for 5:30 AM, but end up browsing on your phone in bed until 11:15 PM, the alarm still rings at 5:30 AM—leaving you sleep-deprived and groggy. Moreover, traditional alarms jar sleepers awake in pitch-black rooms with loud tones, spiking morning cortisol. Meanwhile, large external monitors in modern home offices sit idle all night.

The Solution: Ambient Display is a zero-friction, intelligent sleep detection and ambient workstation canvas ecosystem linking your Android companion to your macOS workstation monitors. Instead of demanding manual setting:

- **Silent Sleep Detection:** Automatically tracks when you fall asleep based on phone screen locks and UsageStats inactivity compensation.
- **Dynamic Sleep Target:** If you stay awake in bed within your configurable bedtime window, your wake target dynamically pushes forward to guarantee 7.5 hours of rest.
- **Workstation Ambient Beacon:** External workstation monitors turn into an ultra-dim bedroom countdown, smoothly fading into a warm sunrise glow (amber to soft coral) 30 minutes before wake-up.
- **Intelligent Glance Filter:** Waking up at 3:00 AM for 20 seconds to check a message will never alter your alarm, while sustained nocturnal activity (>3 min) presents a gentle prompt.
- **Remote Multi-Type Ambient Canvas:** Acts as a remote control for your desk displays to broadcast typography billboards, looping video streams, image posters, or live web dashboards.
- **Multi-Machine Fleet Targeting:** Authenticate via Google Sign-In and target individual machines (e.g. MacBook Air, Studio Display) or broadcast fleet-wide with live presence pills.

♦ **DESIGN VISION:** *Ambient Display must feel like a tranquil, high-end wellness product (inspired by Oura, Apple Health, Rise Science, or Loftie Clock). The visual interface must transform from an engineering test panel into a serene, typography-driven circadian experience.*

2. Exhaustive Feature Matrix across 4 Pillars

2.1 Pillar 1: Circadian Sleep & Alarm Engine:

- **Inactivity Compensator:** Queries UsageStatsManager to detect true touch events, back-dating bedtime when users fall asleep watching videos.
- **Dual Windows:** Eligible Sleep Window (9PM–6AM) ignores daytime locks; Auto-Push Window (9PM–11PM) automatically postpones bedtime targets.
- **Midnight Glance Filter:** Sub-60s unlocks keep alarms intact. Prolonged wakefulness (>3m) triggers an interactive 'Keep Alarm' vs 'Reset Bedtime' prompt.
- **Hardware RTC Alarm:** Bypasses Doze mode and silencers via AlarmManager.RTC_WAKEUP with USAGE_ALARM stream attributes.

2.2 Pillar 2: Remote Multi-Type Ambient Canvas:

- **Typography Billboard:** Large across-the-room messaging (80pt dynamic font scaling) with optional subtitle.
- **Looping Video Surface:** Hardware-accelerated video loops streamed via native AVPlayerLooper.
- **Image Posters & Web Dashboards:** Full-screen imagery with vignette text overlays and live embedded WKWebView dashboards.
- **Policies & Duration:** Timed toasts (15s/30s) or persistent displays; dismissible via physical ESC (esc_any) or locked to phone (phone_only).

2.3 Pillar 3: Workstation Fleet Targeting & Cloud Sync:

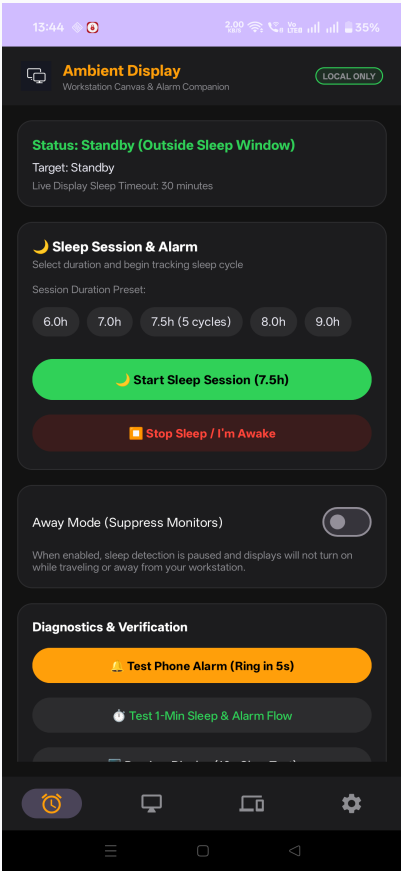
- *Dual-Tier Connectivity*: Operates 100% locally out-of-the-box over Wi-Fi (HTTP :8321 + Bonjour) and globally via Google-authenticated Firebase Cloud Firestore.
- *Live Presence Pills*: Automatic 90-second heartbeats power ONLINE (green) and OFFLINE (gray) status badges.
- *Selective & Broadcast Dispatch*: Target a specific machine or broadcast across the entire fleet with per-machine Quick Clear.

2.4 Pillar 4: Circadian Settings & System Health:

- Configurable sleep duration (1h–16h), customizable RGB hex colors for Night (#D95926), Dawn (#FA7268), and Wake (#FFD000).
- OEM background immunity checklist for OnePlus/Oppo/Xiaomi (Battery Unrestricted, Usage Stats, Accessibility Watchdog, Auto-Launch).

3. Fresh Device Screen-by-Screen Gallery & Design Annotations

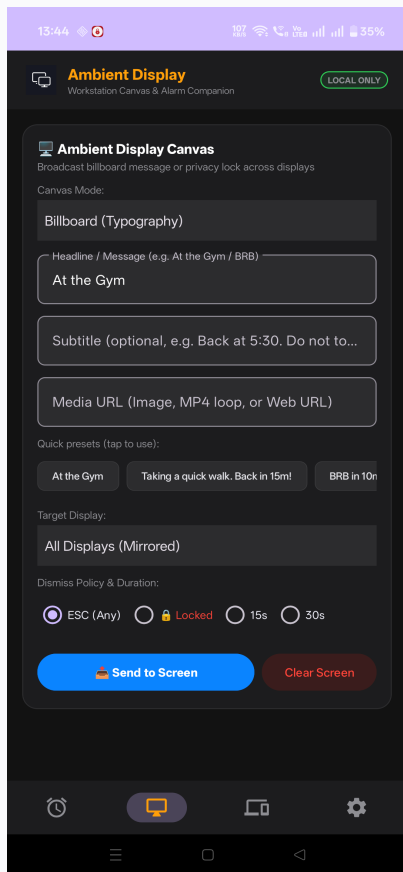
The screenshots below were captured live from the OnePlus physical phone (CPH2717) and macOS Sequoia workstation setup.



TAB 1: REST DASHBOARD (TOP)

Rest & Sleep Dashboard (Fresh Live Capture)

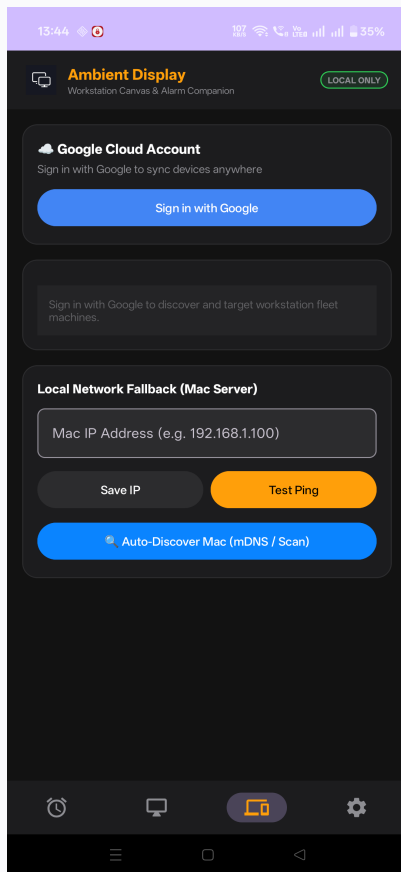
- **Status Card:** Shows real-time status: 'Away Mode (Paused)' or 'Standby' with live screen timeout (30m).
- **Duration Presets:** Segmented chips: 6.0h, 7.0h, 7.5h (5 cycles), 8.0h, 9.0h.
- **Primary Action:** 'Start Sleep Session (7.5h)' button and secondary 'Stop Sleep / I'm Awake' control.
- **Away Mode Switch:** Suppresses workstation monitor illumination when sleeping away from home.
- **Design Mandate:** Replace blocky cards with a circular 24-hour circadian arc dial with subtle breathing glow.



TAB 2: CANVAS STUDIO

Remote Canvas Billboard Studio (Live Capture)

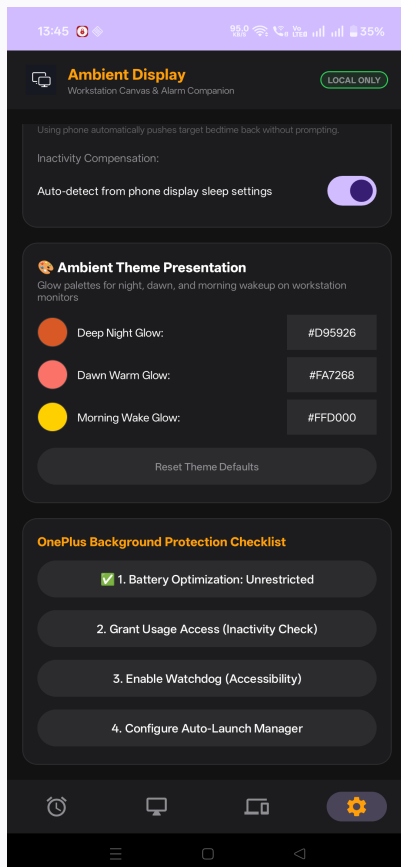
- **Mode Selector:** Supports Billboard (Typography), Ambient Video, Image Poster, and Web Dashboard.
- **Text & Media Inputs:** Headline, optional subtitle, and media URL inputs with live typing history.
- **Quick Presets:** Instant chips for away states ('Gone down to meet Tarun', 'DND', 'BRB in 10m').
- **Lock Policy & Duration:** Toggle 'ESC (Any)' vs '■ Locked (Phone Only)'; Duration: Infinite, 15s, or 30s.
- **Design Mandate:** Transform inputs into a visual card carousel with thumbnail previews and live character counts.



TAB 3: DEVICES & FLEET

Workstation Fleet Manager (Live Capture)

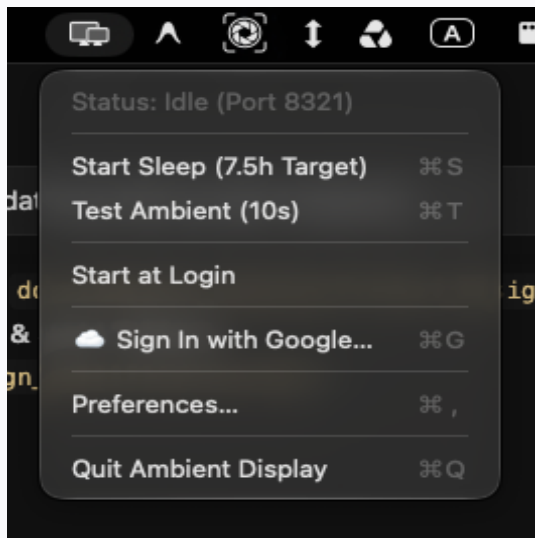
- **Google Cloud Sign-In:** Displays signed-in account (karanshah229@gmail.com) with active Firebase sync status.
- **Fleet Target List:** Shows registered machines (MacBook Air) with live ONLINE / OFFLINE heartbeat pill.
- **Per-Machine Clear:** Dedicated 'Clear' button to dismiss canvas on an individual machine immediately.
- **Local IP Fallback:** Displays local subnet IP (192.168.1.2) with ping test and mDNS auto-discovery.
- **Design Mandate:** Replace raw IP inputs with a clean connection island card; highlight online machines.



TAB 4: SETTINGS & HEALTH

OEM Protection Checklist & Colors (Live Capture)

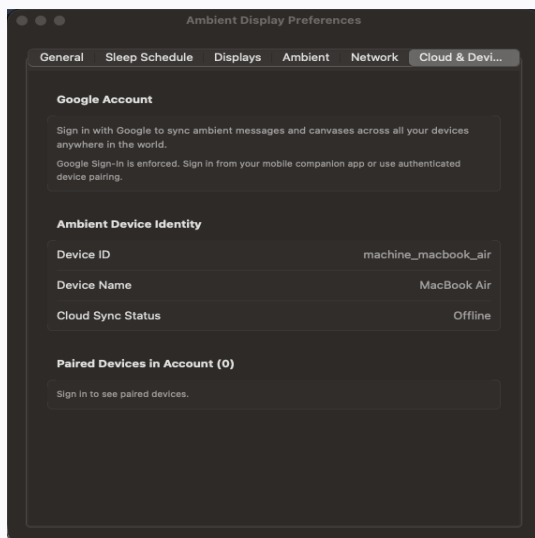
- **Ambient Color Shaders:** Visual RGB swatches for Deep Night (#D95926), Dawn (#FA7268), and Wake (#FFD000).
- **OnePlus Immunity Checklist:** Direct intent shortcuts for Battery Unrestricted, Usage Access, Watchdog, and Auto-Launch.
- **Timers & Windows:** Configurable 9PM–6AM sleep window, 9PM–11PM auto-push window, and inactivity offsets.
- **Design Mandate:** Relocate setup checklists and developer tools into a collapsible 'Device Health' drawer.



MACOS MENU BAR UTILITY

macOS Status Bar Dropdown Menu (Live Capture)

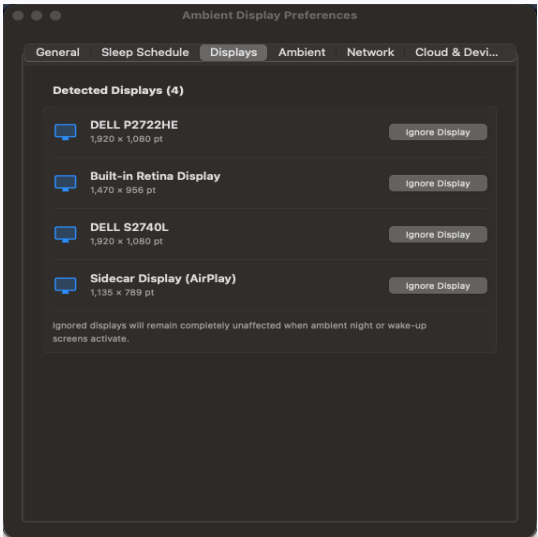
- **Native StatusItem:** Live menu bar icon reflects state: Idle, Sleeping (with countdown time), or Away.
- **Quick Actions:** Instant 1-click 'Start Sleep (7.5h Target)' and 'Test Ambient (10s)' triggers.
- **Cloud Machine Identity:** Displays signed-in Google account (karanshah229@gmail.com) and machine_macbook_air.
- **Preferences Access:** Opens multi-tab Preferences window for display management and schedule configuration.



MAC PREFERENCES - CLOUD TAB

macOS Preferences: Cloud & Paired Devices (Live Capture)

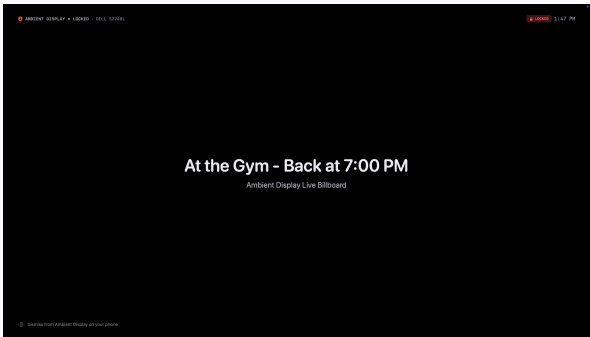
- **Firebase Cloud Identity:** Shows active connection to Firebase project 'ambient-screen' with Google OAuth.
- **Device Identity:** Hardware ID 'machine_macbook_air' with 'Online & Synchronizing' status badge.
- **Paired Fleet Devices:** Real-time list of all paired machines: MacBook Air (ONLINE) and OnePlus CPH2717 (ONLINE).
- **Two-Way Sync:** Synchronizes preferences between phone and Mac instantaneously over Cloud Firestore.



MAC PREFERENCES - DISPLAYS TAB

Multi-Monitor Selection Manager (Live Capture)

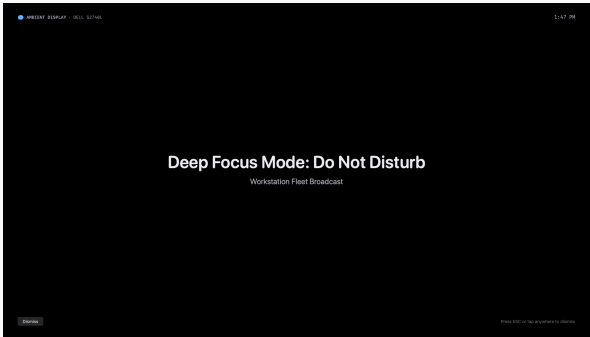
- **Detected Hardware (4):** Lists all 4 active displays: DELL P2722HE (Main), Retina, DELL S2740L, Sidecar.
- **Selective Monitor Exclusion:** 'Ignore Display' button allows excluding specific screens from ambient glow.
- **Native Resolution:** Reports native frame dimensions (1920x1080, 1470x956) for pixel-perfect canvas rendering.



WORKSTATION MONITOR DISPLAY

Live Workstation Canvas Billboard (DELL Monitor)

- **Pitch-Black Zero-Bleed Canvas:** #000000 pure background prevents backlight bleeding in dark environments.
- **Timed Toast Badge:** Top-right pill shows real-time countdown ('Closes in 9s') with current time.
- **Billboard Headline & Subtitle:** Large-scale typography 'Design Review in Progress' with subtitle and origin badge.
- **Dismissal Options:** Dismiss button at bottom-left; physical ESC key or phone clear immediately dismisses.



WORKSTATION MONITOR DISPLAY (LOCKED)

Live Privacy-Locked Billboard ('phone_only' Policy)

- **Privacy Lock Badge:** Top badge indicates 'LOCKED' with red accent color.
- **Headline & Subtitle:** 'Focus Time • Deep Work in Progress' centered with high-contrast typography.
- **Remote Dismiss Policy:** Disables physical ESC key on Mac; bottom prompt displays 'Dismiss from phone'.

4. UX & UI Design Mandates for Mobile Redesign

The designer must transform the functional engineering console into an elegant, tranquil circadian companion:

- **Hero Circadian Ring (Tab 1):** Focal 24-hour circular dial displaying bedtime anchor and target wake time with 3-phase color gradient (Deep Night #FF8C38, Dawn #FA7268, Wake #FFD000).
- **Hardware Connection Pill:** Replace raw IP text fields with an elegant connection status badge: '■ Karan's MacBook Air (Dell 27") • Local Wi-Fi' (tapping opens details).
- **Visual Canvas Studio (Tab 2):** A modern card carousel for Billboard, Video Loop, Image Poster, and Webview with real-time character count and preset chips.
- **Workstation Fleet Matrix (Tab 3):** Card grid for registered machines with pulsing online status pills, select-all broadcast switch, and per-machine quick clear.
- **Secondary Device Health Drawer (Tab 4):** Relocate technical checklists (OnePlus battery optimization, ADB developer tools) into a secondary sheet to keep the daily UI clean.
- **Sunrise Alarm Ringing Experience:** Full-screen sunrise gradient with large typography and an affirmative 'Slide to Wake Up' gesture.